CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 10: Circles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. How many tangents can a circle have? How many of them can be drawn from a point outside the circle?

2. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at Q, with $OQ = 13$ cm. Find PQ.

3. The length of a tangent from a point 10 cm from the centre of a circle is 8 cm. Find the radius.

4. Two concentric circles have radii 10 cm and 6 cm. Find the length of the chord of the larger circle that touches the smaller circle.

5. PA and PB are tangents to a circle with centre O. If $\angle APB = 70^\circ$, find $\angle POA$.

6. Tangents PA and PB from a point P are inclined to each other at 60°, and the radius of the circle is 5 cm. Find the length of each tangent.

7. A quadrilateral $ABCD$ circumscribes a circle. If $AB = 7$ cm, $BC = 10$ cm and $CD = 9$ cm, find AD.

8. The incircle of $\triangle ABC$ touches BC, CA, AB at D, E, F. If $AF = 4$ cm, $BD = 3$ cm and $CE = 5$ cm, find the perimeter of the triangle.

9. Prove that the perpendicular at the point of contact to a tangent of a circle passes through the centre.

10. Prove that a parallelogram circumscribing a circle is a rhombus.

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Answer Key

1. Infinitely many; exactly 2 from an outside point
2. 12 cm
3. 6 cm
4. 16 cm
5. $\angle AOB = 110^\circ$, so $\angle POA = 55^\circ$
6. $\angle APO = 30^\circ$, so $PA = \dfrac{5}{\tan 30^\circ} = 5\sqrt{3}$ cm
7. 6 cm
8. $2(4 + 3 + 5) = 24$ cm
9. The radius at the point of contact is perpendicular to the tangent, and only one perpendicular can be drawn to a line at a given point, so the perpendicular is the line containing the radius.
10. $AB + CD = AD + BC$ and $AB = CD$, $AD = BC$ give $2AB = 2AD$, so all four sides are equal.

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